Structural Admissibility in Rocq

Warrant Debt, Observational Quotients, and Type-Theoretic Verification

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Abstract

Verification can fail before proof search begins. When the observation map of a verification system conflates states that a correctness predicate must distinguish, the verification question is malformed: the required information has already been destroyed by the observational quotient.

We formalise this condition as admissibility. A predicate is admissible exactly when it factors through the equivalence relation induced by observation, or equivalently when the observation-induced quotient is fine enough to preserve the distinctions the predicate requires.

For perturbation systems, we introduce warrant debt as the operational witness of admissibility failure: warrant debt occurs when algebraic observation treats two perturbations as equivalent while empirical evidence distinguishes them. The Warrant Debt Theorem is proved in Rocq: whenever warrant debt exists, the corresponding empirical tolerance predicate is not admissible.

The mechanisation shows that inadmissibility is a prior proof obligation rather than a failure of proof search: verification must establish that the correctness predicate survives quotienting before proof search begins. The Rocq development fully proves the Warrant Debt Theorem, the monitor-as-evidence and conditional safety results, the concrete rupture decomposition theorem for the `SimpleSynergyCompose` instance, and the Non-Factorisation Theorem. The abstract composition theorem is stated at the interface level and is provable only for a concrete composition instance satisfying the required axioms. A higher-order extension to finite dependency families is presented as a proposed generalisation and is not part of the present mechanisation.

Keywords: Formal verification, Observational equivalence, Quotient semantics, Admissibility, Type theory, Rocq.

Contents

1. Introduction

The question is not whether a correctness predicate is true, but whether truth survives quotienting by the observation structure.

Verification failure is commonly diagnosed as proof failure: the prover did not find a certificate, the invariant was too weak, or the abstraction was too coarse. This diagnosis is often wrong.

Many failures are structural: the observation map has already collapsed the distinctions the correctness predicate requires before any proof attempt begins. In that situation, proof search is not merely difficult. The question it is attempting to answer is malformed.

When an observation map $M : X \to O$ is not injective, distinct states share an observational image. The induced equivalence

$$x \sim_M y \iff M(x) = M(y)$$

partitions the state space, and a predicate

$$\Phi : X \to \{0, 1\}$$

is *admissible* with respect to M if and only if it factors through that partition:

$$\Phi = \hat{\Phi} \circ M$$

for some

$$\hat{\Phi} : O \to \{0, 1\}.$$

If Φ is not admissible, there exist states x, y such that $M(x) = M(y)$ and $\Phi(x) \neq \Phi(y)$. No reasoning that operates within the observational regime can certify Φ, because the regime has erased the distinction Φ requires. Inadmissibility is structural impossibility, not epistemic limitation.

The problem belongs to quotient semantics, abstract interpretation, observational equivalence, and definability theory. The issue is not the expressive weakness of the prover; it is whether the observational quotient is fine enough to preserve the distinctions the predicate demands.

Triple-overlap obstruction in regional algebra provides a concrete instance of inadmissibility: the non-vanishing Čech class is the witness that strict associativity is not representable within the existing gluing quotient.

In perturbation systems, this failure appears as *warrant debt*: algebraic observation treats two perturbations as equivalent while empirical evidence distinguishes them. Warrant debt is therefore the concrete witness that admissibility has failed.

Dependent type theory makes admissibility constraints explicit as proof obligations prior to theorem proving. The question is not whether a proof can be found, but whether the correctness predicate is even well-formed relative to the observation structure. The Rocq development presented here makes this precise: warrant debt is represented as a formal object, and the theorem that its existence implies non-admissibility is fully proved.

Contributions. Three results are established. The Factorisation Theorem (Section 2) characterises admissibility in three equivalent forms: factorisation through the observation map, constancy on observation-induced equivalence classes, and refinement of the correctness partition by the observational partition. It is the formal criterion against which all subsequent results are measured. The Warrant Debt Theorem (Section 4) proves that whenever empirical evidence distinguishes algebraically equivalent perturbations, the corresponding empirical tolerance predicate is not admissible; warrant debt is thus the operational witness of admissibility failure, and the theorem is fully mechanised in Rocq. The mechanisation (Sections 5–6) exhibits admissibility checking as a prior proof obligation preceding monitor correctness and compositional reasoning. A concrete rupture decomposition theorem is proved for the `SimpleSynergyCompose` instance, supplying the binary compositional witness, and the Non-Factorisation Theorem is established in the Rocq development. A higher-order extension to finite dependency families on n components is set out at the level of mathematical structure as a proposed generalisation; minimal support sets identify the smallest subsystems on which the global predicate genuinely depends. The higher-order theorem is not part of the present mechanisation.

Scope. This is a formal verification paper. Motivation is provided only where it identifies a formal problem. Metaphysical interpretation belongs to a separate paper in this programme.

2. Observational Quotients and Admissibility

This section states the abstract theorem of which all subsequent results are instances. No perturbation-specific machinery is required here.

2.1. Observation Maps and Observational Equivalence

Let X be a non-empty set (the state space) and O a non-empty set (the observation space). Let

$$M : X \to O$$

be a function, the *observation map*; M need not be injective.

Definition 2.1 (Observational Equivalence)**.** The *observational equivalence* induced by M is

$$x \sim_M y \iff^{\text{def}} M(x) = M(y).$$

This is an equivalence relation on X.

Let $X/{\sim_M}$ denote the quotient and

$$\pi : X \to X/{\sim_M}$$

the canonical projection.

There exists a unique injection

$$\bar{M} : X/{\sim_M} \to O$$

such that

$$M = \bar{M} \circ \pi.$$

The observational equivalence is the relevant object of study. Two states that M cannot separate are, from the perspective of the verification system, identical. No predicate that respects the observation structure can distinguish them.

2.2. Admissible Predicates

Definition 2.2 (Admissibility)**.** Let

$$\Phi : X \to \{0, 1\}$$

be a correctness predicate.

Φ is *admissible with respect to* M if and only if there exists

$$\widetilde{\Phi} : X/{\sim_M} \to \{0, 1\}$$

such that

$$\Phi = \widetilde{\Phi} \circ \pi.$$

That is, Φ must factor through the quotient induced by observational equivalence.

Equivalently, since

$$M = \bar{M} \circ \pi,$$

admissibility is equivalent to factorisation through the observation map itself: there exists

$$\hat{\Phi} : O \to \{0, 1\}$$

such that

$$\Phi = \hat{\Phi} \circ M.$$

The quotient is primary. The codomain O matters only because the observation map induces the equivalence relation that defines the quotient. Verification is therefore governed not by the observation values themselves, but by the partition they impose on the state space.

2.3. The Factorisation Theorem

Theorem 2.3 (Factorisation)**.** *The following conditions are equivalent.*

(1) Φ is admissible with respect to M.
(2) Φ is constant on $\sim_M$-equivalence classes:

$$M(x) = M(y) \;\Rightarrow\; \Phi(x) = \Phi(y).$$

(3) The observational equivalence refines the predicate equivalence:

$$\sim_M \subseteq \sim_\Phi,$$

where

$$x \sim_\Phi y \quad \overset{\text{def}}{\iff} \quad \Phi(x) = \Phi(y).$$

Proof. (1) $\Rightarrow$ (2): If

$$\Phi = \hat{\Phi} \circ M$$

and

$$M(x) = M(y),$$

then

$$\Phi(x) = \hat{\Phi}(M(x)) = \hat{\Phi}(M(y)) = \Phi(y).$$

(2) $\Rightarrow$ (3): If

$$x \sim_M y,$$

then

$$M(x) = M(y),$$

so by (2),

$$\Phi(x) = \Phi(y),$$

hence

$$x \sim_\Phi y.$$

Therefore

$$\sim_M \subseteq \sim_\Phi .$$

(3) $\Rightarrow$ (1): Since

$$\sim_M \subseteq \sim_\Phi,$$

the assignment

$$\widetilde{\Phi}([x]_M) = \Phi(x)$$

is well-defined on the quotient

$$X/{\sim_M}.$$

Thus

$$\Phi = \widetilde{\Phi} \circ \pi.$$

Extending $\widetilde{\Phi}$ along $\bar{M}$ yields the equivalent factorisation through O. $\square$

Theorem 2.3 is the spine of the paper.

A predicate is decidable from observation if and only if the observation-induced partition is fine enough to respect the predicate's partition.

If the observational equivalence is too coarse, no argument about the predicate can recover what the quotient has already destroyed.

2.4. Saturation

Definition 2.4 (Saturation)**.** A set

$$S \subseteq X$$

is *M-saturated* if for all

$$x \in S \quad \text{and} \quad y \in X,$$

$$M(x) = M(y) \;\Rightarrow\; y \in S.$$

That is, S is a union of $\sim_M$-equivalence classes.

Corollary 2.5 (Saturation Criterion)**.** *Φ is admissible with respect to M if and only if*

$$\Phi^{-1}(1)$$

is M-saturated.

Proof. Admissibility holds if and only if no $\sim_M$-equivalence class straddles the boundary between

$$\Phi^{-1}(1) \quad \text{and} \quad \Phi^{-1}(0).$$

This holds if and only if

$$\Phi^{-1}(1)$$

is a union of $\sim_M$-classes, which is exactly the definition of M-saturation. □

2.5. Inadmissibility as Structural Impossibility

When Φ is not admissible, there exist

$$x, y \in X$$

such that

$$M(x) = M(y) \quad \text{and} \quad \Phi(x) \neq \Phi(y).$$

The pair (x, y) is an inadmissibility witness.

No predicate that factors through the current observation map can recover this distinction while that observational regime remains fixed. The quotient

$$X/{\sim_M}$$

has already collapsed the distinction that Φ requires.

Inadmissibility is not epistemic ignorance. It is structural impossibility.

The failure lies in the representational regime itself, not in the strength of the prover operating within it.

2.6. Literature Placement

Theorem 2.3 is a definability result. Its connections to the literature are structural.

In abstract interpretation [7, 8], a Galois connection (α, γ) relates concrete and abstract domains; a property survives abstraction only if the abstract domain is fine enough to preserve the required distinctions [9, 13]. The quotient $X/{\sim_M}$ is the abstract domain. Inadmissibility is the case in which the abstraction is too coarse and the required distinction has been irreversibly erased.

Beth's theorem [4] establishes that implicit and explicit definability coincide in first-order logic. In the present setting there is a fixed observation map rather than a first-order theory, and the observation-induced equivalence plays the role of the theory's indistinguishability relation. A predicate is definable in the relevant sense if and only if the quotient is fine enough to preserve its partition [16].

Two structures are elementarily equivalent if and only if the Duplicator wins the Ehrenfeucht–Fraïssé game of all finite lengths [11, 12]. The relation $\sim_M$ is the observational analogue: two states are $\sim_M$-equivalent when no observation separates them. Inadmissibility is the corresponding structural result: collapse of the observational quotient forces indistinguishability between states that the correctness predicate must separate [18].

Milner's observational equivalence [19, 20] identifies processes that no context can distinguish by observable actions. Observation maps play the role of contexts, and the admissibility condition is the verification analogue of contextual equivalence [25].

3. Concrete Perturbation Space

The abstract framework of Section 2 is instantiated here.

The observation map M becomes an algebraic observation map over perturbations: it preserves carrier class and magnitude while discarding emergent boundary information.

The correctness predicates Φ become tolerance predicates over interface regions.

This section introduces only the formal objects required for the Warrant Debt Theorem of Section 4; nothing here is independent of that purpose.

3.1. Boundary Type Classification

Definition 3.1 (Boundary Types)**.** A perturbation is classified by one of three boundary types.

(i) `INV` (invariant): tolerance conditions are fully determined at design time by algebraic structure;
(ii) `REL` (relational): tolerance conditions vary within bounded and certifiable limits;
(iii) `EMG` (emergent): tolerance conditions depend on information that cannot be recovered from the algebraic observation kernel alone.

The distinction is representational rather than merely descriptive: `EMG` marks precisely the case in which algebraic observation is structurally insufficient.

```
Inductive BType : Type :=
  | INV  (* invariant: structurally determined *)
  | REL  (* relational: bounded, variable      *)
  | EMG. (* emergent: observationally lost     *)

Record Perturbation := mkPerturbation {
  p_btype     : BType;
  p_carrier   : CarrierClass;
  p_magnitude : Interval;
  p_mode      : nat
}.
```

Listing 1: Boundary types and perturbation record.

The `Interval` type carries a proof of well-formedness $\ell \leq h$, so every magnitude is represented as a certified interval rather than an uncertified real pair.

A structural consequence used later is that `EMG` is the top element under composition: if either component is emergent, the composition is emergent. Composition cannot eliminate emergent character.

3.2. Tolerance Predicates

Definition 3.2 (Tolerance Predicates)**.** A tolerance predicate specifies whether an interface region can absorb a perturbation without violating its safety boundary. Two forms are distinguished: formal tolerance, decided at design time by a Boolean procedure; and empirical tolerance, indexed by evidence such as sealed monitoring logs or certified test records.

```
Definition TolPredicate := Perturbation -> Prop.

Definition formal_tol (dec : Perturbation -> bool)
  : TolPredicate :=
  fun p => dec p = true.

Parameter Evidence : Type.
Parameter empirical_tol : Evidence -> TolPredicate.
```

Listing 2: Tolerance predicates.

The type `Evidence` is left abstract. The results of Section 4 do not depend on the internal structure of evidence, only on the fact that empirical tolerance may distinguish perturbations that algebraic observation treats as equivalent.

This gap is the source of warrant debt.

3.3. Rupture

Definition 3.3 (Rupture)**.** A perturbation p *ruptures* tolerance predicate `tol` when

$$\neg\,\mathtt{tol}(p).$$

```
Definition Rupture
  (tol : TolPredicate)
  (p : Perturbation) : Prop :=
  ~ tol p.
```

Listing 3: Rupture.

Rupture is not a property of the perturbation alone. It is a relation between a perturbation and a tolerance predicate.

Changing the predicate changes what counts as rupture.

This matters because detectability depends not only on whether rupture occurs, but on whether the tolerance predicate itself is admissible with respect to the observation structure. Section 4 makes this precise.

4. Algebraic Observation and Warrant Debt

This section is the formal centre of the paper.

Section 2 established the abstract admissibility criterion: a predicate is admissible if and only if the observational equivalence induced by the observation map is fine enough to preserve the distinctions the predicate requires.

This section is the concrete instance of that criterion for perturbation systems.

The algebraic observation map plays the role of M; `alg_equiv` plays the role of $\sim_M$; and `PredAdmissible` is the perturbation-level factorisation condition of Theorem 2.3.

4.1. The Algebraic Observation Map

The algebraic observation map strips emergent boundary information.

Two perturbations are algebraically equivalent when they agree on carrier class and magnitude and both belong to the algebraically observable fragment of the system.

```
Definition btype_algebraic (b : BType) : bool :=
  match b with
  | INV => true
  | REL => true
  | EMG => false
  end.

Definition alg_equiv (p q : Perturbation) : Prop :=
  p_carrier p   = p_carrier q /\
  p_magnitude p = p_magnitude q /\
  btype_algebraic (p_btype p) = true /\
  btype_algebraic (p_btype q) = true.

Definition PredAdmissible
  (P : Perturbation -> Prop) : Prop :=
  forall p q : Perturbation,
    alg_equiv p q -> (P p <-> P q).
```

Listing 4: Algebraic equivalence and predicate admissibility.

Definition 4.1 (Observational Admissibility in Perturbation Space)**.** A predicate P is *admissible* with respect to the algebraic observation map if and only if it is invariant under `alg_equiv`: P cannot distinguish perturbations that the algebraic kernel cannot distinguish.

This is the perturbation-level realisation of condition (2) of Theorem 2.3:

$$M(x) = M(y) \;\Rightarrow\; \Phi(x) = \Phi(y).$$

Emergent perturbations lie outside the algebraic observation domain entirely:

$$\texttt{btype\_algebraic(p\_btype q) = false}$$

whenever

$$\texttt{p\_btype q = EMG}.$$

The algebraic kernel therefore forms no equivalence class containing an emergent perturbation.

Any predicate that separates an algebraic perturbation from an emergent one witnesses information that the kernel destroys.

4.2. Warrant Equivalence

Definition 4.2 (Warrant Equivalence)**.** Perturbations p and q are *warrant-equivalent* with respect to evidence e when empirical tolerance treats them identically:

$$\texttt{warrant\_equiv}(e, p, q) \overset{\text{def}}{\iff} \Big(\texttt{empirical\_tol}(e, p) \leftrightarrow \texttt{empirical\_tol}(e, q)\Big).$$

```
Definition warrant_equiv
  (e : Evidence)
  (p q : Perturbation) : Prop :=
  empirical_tol e p <-> empirical_tol e q.
```

Listing 5: Warrant equivalence.

Warrant equivalence is the empirical counterpart of algebraic equivalence.

Algebraic equivalence records what the formal certification infrastructure treats as the same perturbation.

Warrant equivalence records what the empirical evidence treats as the same perturbation.

Warrant debt arises precisely when these two notions of sameness disagree.

4.3. Warrant Debt

Definition 4.3 (Warrant Debt)**.** A *warrant debt* exists between perturbations p and q with respect to evidence e when they are algebraically equivalent but not warrant-equivalent:

$$\mathrm{WarrantDebt}(e, p, q) \overset{\text{def}}{\iff} \texttt{alg\_equiv}(p, q) \wedge \neg\texttt{warrant\_equiv}(e, p, q).$$

```
Definition WarrantDebt
  (e : Evidence)
  (p q : Perturbation) : Prop :=
  alg_equiv p q /\ ~ warrant_equiv e p q.
```

Listing 6: Warrant debt.

A $\mathrm{WarrantDebt}(e, p, q)$ is the formal witness that the observational regime is too coarse.

The algebraic certification infrastructure treats p and q as equivalent, while the evidence record distinguishes them.

Admissibility therefore fails at the level of representation itself: the system is attempting to certify a predicate that its own observation structure cannot support.

4.4. The Warrant Debt Theorem

Theorem 4.4 (Warrant Debt Implies Non-Admissibility)**.** *For any evidence e and perturbations p and q,*

$$\mathrm{WarrantDebt}(e, p, q) \Rightarrow \neg\texttt{PredAdmissible}\big(\texttt{empirical\_tol}(e)\big).$$

Proof. Suppose

$$\mathrm{WarrantDebt}(e, p, q).$$

Then

$$\texttt{alg\_equiv}(p, q)$$

and

$$\neg\texttt{warrant\_equiv}(e, p, q).$$

Assume for contradiction that

$$\texttt{PredAdmissible}\big(\texttt{empirical\_tol}(e)\big).$$

By Definition 4.1, admissibility requires that algebraically equivalent perturbations cannot be separated by the predicate.

Hence

$$\texttt{alg\_equiv}(p, q)$$

implies

$$\texttt{empirical\_tol}(e, p) \leftrightarrow \texttt{empirical\_tol}(e, q),$$

which is exactly

$$\texttt{warrant\_equiv}(e, p, q).$$

This is the contradiction. Admissibility forces warrant equivalence, while warrant debt asserts its negation. Both cannot hold simultaneously.

□

The Rocq mechanisation is fully proved:

```
Theorem warrant_debt_implies_non_admissible :
  forall (e : Evidence)
         (p q : Perturbation),
  WarrantDebt e p q ->
  ~ PredAdmissible (empirical_tol e).
Proof.
  intros e p q [Hequiv Hne_warrant] Hadm.
  apply Hne_warrant.
  unfold PredAdmissible in Hadm.
  exact (Hadm p q Hequiv).
Qed.
```

Listing 7: Warrant Debt Theorem, fully proved in Rocq.

The proof is four lines. Its brevity reflects the precision of the definitions: the mathematics is in the definitions, not in the proof scripts. A correct formalisation makes the theorem transparent; a bloated one obscures it.

The engineering consequence is immediate. If a warrant debt can be exhibited, the system is operating with an inadmissible observational regime. More proof cannot repair representational failure. The correct response is not stronger proof search, but refinement of the observation map until admissibility is restored, or formal recognition that the target predicate is not verifiable within the current regime.

Connection to Theorem 2.3. The Warrant Debt Theorem is the concrete instance of the Factorisation Theorem for perturbation systems.

`alg_equiv` is the induced equivalence $\sim_M$.

`PredAdmissible` is the factorisation condition.

The pair (p, q) with $\text{WarrantDebt}(e, p, q)$ is the inadmissibility witness of Theorem 2.3: two states the observation map conflates that the predicate must separate.

The implication of Theorem 4.4 is the perturbation-level form of condition (2) in Theorem 2.3.

These are not two theorems. They are one theorem at two levels of abstraction.

5. Type-Level Realisation

This section shows how admissibility appears as a proof obligation inside dependent type theory.

The type structure does not replace the mathematics; it exhibits the same obligations in machine-checkable form. Refinement of observation corresponds to refinement of the formal objects through which verification proceeds.

Theorem 4.4 established that warrant debt implies non-admissibility. The present section states the operational consequence: before proving a predicate, one must first establish that the predicate is well-formed relative to the available observation structure.

5.1. Observational Refinement as Product Extension

When a predicate Φ is inadmissible with respect to an observation map M_i, the corrective is not stronger proof search but observational refinement: introduce an auxiliary observation that separates the inadmissibility witnesses.

Definition 5.1 (Observational Refinement)**.** Let

$$M_i : X \to O_i.$$

An *observational refinement* of M_i is a map

$$M_{i+1} : X \to O_{i+1}$$

such that

$$x \sim_{M_{i+1}} y \;\Rightarrow\; x \sim_{M_i} y.$$

That is, the refined observation induces a strictly finer partition. Equivalently,

$$M_{i+1} = (M_i, N_i)$$

for some auxiliary observation

$$N_i : X \to F_i,$$

so that

$$O_{i+1} \cong O_i \times F_i.$$

Proposition 5.2 (Product Decomposition)**.** *If*

$$M_{i+1} = (M_i, N_i)$$

is an observational refinement, then

$$O_{i+1} \cong O_i \times F_i,$$

and the projection

$$\pi_1 : O_{i+1} \to O_i$$

satisfies

$$M_i = \pi_1 \circ M_{i+1}.$$

The refined observation reduces to the original by projection.

Each refinement step therefore extends the type of the observation space by a product factor. The projection π_1 corresponds exactly to forgetting the new factor.

Refinement is not replacement. It is conservative extension: the old observation remains recoverable by projection, while the new factor provides the separation required to eliminate inadmissibility.

5.2. Type Structure and Proof Obligations

Type structure enforces the projection and refinement obligations corresponding to Theorem 2.3.

The issue is not whether a proof can be found, but whether the predicate is even well-formed relative to the observation structure.

A predicate admissible with respect to M_i can be stated without appeal to distinctions outside the partition induced by M_i. A non-admissible predicate cannot: any attempt to verify it requires appealing to a finer partition, which is exactly the formal signal that the observation map must be refined.

In the Rocq development, `PredAdmissible`(P) is the explicit proof obligation that must be discharged before verification of a tolerance predicate may proceed. If it cannot be discharged, the observation map must be refined by product extension as in Proposition 5.2, or it must be formally certified that P is not verifiable within the current regime. Only after admissibility is established does proof search for P begin.

This ordering is not a recommendation about proof strategy. It is a structural consequence of Theorem 4.4.

Warrant debt proves that proof search inside an inadmissible observational regime is not incomplete but misdirected: the system is attempting to certify a predicate its own representation cannot support.

6. Composition and Rupture Decomposition

6.1. Composition Signature

Composition is expressed as a Rocq module type. A compliant operator must satisfy three axioms: emergent type propagation, magnitude monotonicity, and a bridge between composite rupture and its sources.

```
Module Type COMPOSITION_SIG.
  Parameter par_compose :
    Perturbation -> Perturbation -> Perturbation.

  Parameter synergy_debt :
    Perturbation -> Perturbation -> Prop.

  Axiom compose_btype_emg :
    forall p q,
    p_btype p = EMG \/ p_btype q = EMG ->
    p_btype (par_compose p q) = EMG.

  Axiom compose_magnitude_ub :
    forall p q,
      (iv_hi (p_magnitude p) <= iv_hi (p_magnitude (par_compose p q)))%
         R /\
      (iv_hi (p_magnitude q) <= iv_hi (p_magnitude (par_compose p q)))%
         R.

  Axiom rupture_synergy_bridge :
    forall (tol : TolPredicate) (p q : Perturbation),
    Rupture tol (par_compose p q) ->
    Rupture tol p \/ Rupture tol q \/ synergy_debt p q.
End COMPOSITION_SIG.
```

Listing 8: Abstract composition module signature.

Definition 6.1 (Interaction Debt, Binary Case)**.** Two perturbations incur *interaction debt* (in this binary instance also called *synergy debt*) when both are individually tolerated under a predicate, yet their composition ruptures under that same predicate. The binary case names the lowest non-trivial level of the interaction debt hierarchy and is not reducible to component-level failure. The general hierarchy, which classifies interaction debt by the size of the minimal support set on which inadmissibility arises, is developed in Section 6.3.

6.2. Proved Rupture Decomposition

The concrete module `SimpleSynergyCompose` satisfies the axioms of `COMPOSITION_SIG` with full proofs. The rupture decomposition is therefore established constructively.

Theorem 6.2 (Rupture Decomposition)**.** *For any tolerance predicate* `tol` *and perturbations p, q,*

$$\mathrm{Rup}(\texttt{tol}, \texttt{par\_compose}(p,q)) \;\Rightarrow\; \mathrm{Rup}(\texttt{tol}, p) \;\vee\; \mathrm{Rup}(\texttt{tol}, q) \;\vee\; \texttt{synergy\_debt}(p,q).$$

```
Lemma rupture_synergy_bridge :
  forall (tol : TolPredicate) (p q : Perturbation),
```

```
  Rupture tol (par_compose p q) ->
  Rupture tol p \/ Rupture tol q \/ synergy_debt p q.
Proof.
  intros tol p q Hrup.
  destruct (classic (tol p)) as [Hp | Hnp].
  - destruct (classic (tol q)) as [Hq | Hnq].
    + right. right. unfold synergy_debt. exists tol.
      exact (conj Hp (conj Hq Hrup)).
    + right. left. exact Hnq.
  - left. exact Hnp.
Qed.

Theorem rupture_decomposition_proved :
  forall (tol : TolPredicate) (p q : Perturbation),
  Rupture tol (SimpleSynergyCompose.par_compose p q) ->
  Rupture tol p \/ Rupture tol q \/
  (exists tol' : TolPredicate,
    tol' p /\ tol' q /\
    Rupture tol' (SimpleSynergyCompose.par_compose p q)).
Proof.
  intros tol p q Hrup.
  exact (SimpleSynergyCompose.rupture_synergy_bridge tol p q Hrup).
Qed.
```

Listing 9: Proved rupture decomposition.

The third disjunct identifies the interaction term: both components satisfy a tolerance predicate, yet their composition violates it. Interaction debt at the binary level is therefore not residual; it is a named and typed obligation.

The role of Theorem 6.2 is compositional. Where Theorem 4.4 identifies failure of admissibility at the level of a single predicate, Theorem 6.2 identifies the sources of failure in composed systems for the binary case. A composite rupture on two components decomposes into certified local failures or a single binary interaction term. The binary formulation is, however, only the first non-trivial case of a hierarchy. For systems with three or more components, a single residual interaction term is no longer sufficient: it records that some debt exists without identifying the minimal locus of failure. Section 6.3 develops the corresponding hierarchy.

6.3. Granular Synergy and Higher-Order Warrant Debt

The binary formulation of rupture decomposition is only the first non-trivial case. In general, for a system with component family $C = \{C_1, \dots, C_n\}$, global inadmissibility cannot be reduced to a single residual interaction term without loss of explanatory force. Such a residual merely records that some interaction debt exists; it does not identify the minimal locus of failure.

The correct generalisation is hierarchical. For each non-empty subset $S \subseteq C$, let Σ_S denote the warrant debt associated with the interaction structure of the components in S, namely the failure of the global predicate to factor through the observation map restricted to that subsystem. Then local failure is the case $|S| = 1$, pairwise interaction debt is the case $|S| = 2$ recovered by Theorem 6.2, triple interaction debt the case $|S| = 3$, and so forth.

Accordingly, global rupture admits the decomposition

$$\Phi_{\text{global}} \;=\; \bigvee_{S \subseteq C,\, S \neq \varnothing} \Sigma_S,$$

where the operative question is not merely whether some Σ_S is non-zero, but which subsets S are minimal witnesses of inadmissibility.

Definition 6.3 (Dependency Family and Minimal Support)**.** The *dependency family* of a global predicate Φ over the component family C is

$$\mathcal{D}(\Phi) \;=\; \{\, S \subseteq C \mid \Phi \text{ requires } S \text{ for admissibility} \,\}.$$

A subset $S \in \mathcal{D}(\Phi)$ is a *minimal support set* when no proper subset of S lies in $\mathcal{D}(\Phi)$. The minimal elements of $\mathcal{D}(\Phi)$ identify the smallest subsystems on which Φ genuinely depends.

A predicate may be admissible on every proper projection of size less than k, yet fail to be admissible on the full system. This is genuine k-th order synergy, not reducible to lower-order interaction.

Theorem 6.4 (Higher-Order Warrant Debt)**.** *Let Φ be a global predicate on a component family $C = \{C_1, \ldots, C_n\}$ with admissibility-induced dependency family $\mathcal{D}(\Phi)$. If Φ is inadmissible relative to the observation map on C, and if for every proper subset $S' \subsetneq S$ of size strictly less than some $k \leq n$ the projected predicate $\Phi|_{S'}$ is admissible, then there exists a minimal support set $S^* \in \mathcal{D}(\Phi)$ with $|S^*| \geq k$. The set S^* is the smallest subsystem on which higher-order warrant debt arises, and it is the minimal witness of inadmissibility.*

Proof sketch. By the Factorisation Theorem (Section 2), inadmissibility of Φ is equivalent to the existence of a pair of global states identified by the observation map but separated by Φ. Project that pair onto every subset $S \subseteq C$. The collection of subsets on which the projected pair remains separated forms an upward-closed family inside the powerset lattice of C; its minimal elements are the minimal support sets of $\mathcal{D}(\Phi)$. By hypothesis, no subset of size less than k lies in this family, so every minimal element has cardinality at least k. Pick any such minimal element S^*. By construction $\Phi|_{S^*}$ is inadmissible while $\Phi|_{S'}$ is admissible for every $S' \subsetneq S^*$, so S^* is a minimal witness of inadmissibility at order $|S^*|$. $\square$

Remark 6.5 (Mechanisation status). The binary case $k = 2$ is the content of Theorem 6.2 and is fully mechanised in Rocq via `rupture_synergy_bridge` for the concrete `SimpleSynergyCompose` instance. The general case is established here at the level of mathematical structure and is presented as a proposed extension to the present development. A Rocq mechanisation parameterised over the cardinality of C would reduce to the lattice structure of the powerset of C, but it is outside the scope of the present formalisation, which proves only the binary concrete instance.

Remark 6.6 (Significance for industrial systems). The significance is practical as well as formal. In industrial systems, all local indicators may remain within tolerance, and every pairwise relation may appear acceptable, while the joint configuration violates a global invariant. Pressure, temperature, compressor lag, and flow timing may each be locally benign yet jointly inadmissible. Such failures are invisible to anomaly detection understood as local deviation, but they are precisely what admissibility theory detects.

6.4. Bridge Theorem and Audit Interpretation

The bridge from rupture to audit therefore requires classification rather than mere existential detection. The bridge result must identify not only that interaction debt exists, but the minimal support set on which it arises. Only then can verification terminate in assignable warrant debt: which observational boundary failed, which refinement is required, and which institutional responsibility is implicated.

Corollary 6.7 (Strengthened Rupture-to-Synergy Bridge). *If a global predicate Φ on the component family C ruptures, and if no lower-order interaction debt of order strictly less than k suffices to witness this rupture, then the rupture implies the existence of a minimal support set $S^* \in \mathcal{D}(\Phi)$ of order at least k that witnesses higher-order warrant debt.*

This is stronger than the binary bridge axiom of Section 6, which records existential interaction debt without classifying its order. The strengthened formulation converts a rhetorical remainder into a formal object: the minimal witness of inadmissibility, located at a definite level of the dependency hierarchy.

Together, the three results (Theorem 4.4, Theorem 6.2, and Theorem 6.4) separate distinct failure modes. Of these, the first two are fully mechanised in Rocq for the concrete `SimpleSynergyCompose` instance, while the third is established here at the level of mathematical structure and is identified in Section 7 as a proposed extension of the Rocq development. Representational failure, certified by warrant debt, requires refinement of the observation map. Compositional failure, certified at the binary level by rupture decomposition and at higher orders by minimal support sets within the dependency family, is analysed within a fixed observational regime. The distinction is structural: refinement cannot eliminate higher-order warrant debt, and composition cannot repair inadmissibility.

7. Mechanisation Boundary

The principal non-factorisation result is now fully proved in the Rocq development. In particular, the theorem `no_alg_factorisation`, established in `bdgi_perturbation.v` and ported into the canonical development file `bdgi_perturbation_proved.v`, establishes by explicit witness construction that the rupture predicate cannot factor through the algebraic quotient induced by `alg_equiv`. This discharges the earlier conjectural status of non-factorisation and makes precise the claim that observational indistinguishability is a structural rather than merely epistemic limitation.

Theorem 7.1 (Non-Factorisation). *There is no algebraic homomorphism*

$$f : \texttt{Perturbation} \to \texttt{AlgPerturbation}$$

that preserves rupture for every tolerance predicate.

Equivalently, no total projection from the full perturbation space to its algebraically observable fragment can preserve rupture behaviour in general.

```
Definition AlgPerturbation :=
  { p : Perturbation |
    btype_algebraic (p_btype p) = true }.

Definition AlgHom :=
  Perturbation -> AlgPerturbation.

Definition proj_alg (ap : AlgPerturbation) : Perturbation :=
  proj1_sig ap.

Definition RupturePreserving
  (f : AlgHom)
  (tol : TolPredicate) : Prop :=
  forall p : Perturbation,
    Rupture tol p <->
    Rupture tol (proj_alg (f p)).
```

Listing 10: Non-factorisation type signature.

The Rocq proof fixes an arbitrary f : `AlgHom` and an emergent perturbation p_0 with `p_btype`(p_0) = `EMG`, then constructs a witness tolerance predicate that holds on every algebraic perturbation and fails at p_0. This contradicts rupture preservation and discharges the proof obligation explicitly within the canonical development file.

Remark 7.2 (Correction of an earlier formulation). An earlier draft of this development presented a related claim, labelled `emg_sensitive_not_admissible`, asserting that any predicate sensitive to the emergent–algebraic distinction is inadmissible with respect to the algebraic observation map. Under the present definition of `alg_equiv` that formulation is in fact false, and the corresponding lemma has been removed from the development rather than retained as an `Admitted` declaration. The witness-based non-factorisation result stated above is the correct theorem. It captures the substantive content of the earlier claim, namely that the algebraic regime cannot retain the distinctions on which rupture depends, without the spurious universal quantification over admissibility predicates that rendered the earlier formulation unsound.

7.1. The Abstract Composition Boundary

The only remaining admitted theorem in the development is the abstract statement `rupture_decomposition`, formulated over the parameter `par_compose`. This is intentional. Since `par_compose` is introduced as an opaque composition operator, the theorem is not derivable without additional composition axioms. Rather than proving an artefact of an unconstrained abstraction, the development proves the concrete theorem `rupture_decomposition_proved` for the explicit `SimpleSynergyCompose` instance, where the required bridge laws are established.

This distinction is methodological rather than provisional. The abstract statement specifies the interface conditions required of any admissible composition operator, and the concrete theorem certifies one realised instance satisfying those conditions. Any further module bound by `COMPOSITION_SIG` obtains the theorem at instantiation time, conditional on supplying the axioms the signature names.

7.2. Higher-Order Generalisation

Higher-order warrant debt for systems with more than two interacting components is not part of the present mechanisation. The current Rocq development proves the binary case, in which interaction debt first appears non-trivially. The finite dependency lattice required for the general case is mathematically straightforward but proof-theoretically lengthy, and is reserved for subsequent work.

8. Conclusion

Formal verification does not certify truth first. It certifies admissibility first. Only after admissibility is established does proof begin.

The Factorisation Theorem states the abstract criterion: a predicate is decidable from observation if and only if the observation-induced partition is fine enough to preserve the distinctions the predicate requires.

The Warrant Debt Theorem is the concrete operational instance of that criterion. When algebraically equivalent perturbations are distinguished by empirical evidence, the corresponding empirical tolerance predicate is not admissible. This result is fully mechanised in Rocq.

The Rupture Decomposition Theorem extends the account to composed systems. At the binary level, composite failure reduces to one of two cases: local component rupture or a named interaction term. The binary result is fully proved in Rocq for the concrete `SimpleSynergyCompose` instance, and the abstract theorem over `COMPOSITION_SIG` is provable for any module that supplies the required composition axioms.

The higher-order extension generalises this account to the full dependency family. Global failures decompose into local failures, lower-order interaction debts, or irreducible higher-order warrant debt determined by minimal support sets within the dependency family of the global predicate. The minimal support set identifies the smallest subsystem on which the global predicate genuinely depends, and therefore the smallest locus at which institutional responsibility can be assigned. This higher-order extension is presented at the level of mathematical structure and is identified as a proposed extension to the present Rocq development rather than a result of it.

The Non-Factorisation Theorem is fully proved in the Rocq development and rules out any total algebraic projection that preserves rupture behaviour for every tolerance predicate.

Together, these results separate two distinct sources of verification failure. Representational failure occurs when the observational regime is too coarse and admissibility fails; it requires refinement of the observation map. Compositional failure occurs when admissibility holds but interaction produces rupture; it is analysed within the fixed observational regime.

More proof cannot repair the first. More abstraction cannot eliminate the second.

The connections to abstract interpretation [7, 8], Ehrenfeucht–Fraïssé indistinguishability [11, 18], Beth definability [4, 16], and observational equivalence [19, 20] are structural rather than decorative.

They place admissibility inside an established family of results concerning abstraction, quotienting, and definability: when the quotient destroys the distinction, proof cannot restore it.

One further observation.

Whitehead's account of process treats the constitution of a proposition and the determination of its observational conditions as inseparable [31].

The type

$$\texttt{WarrantDebt}(e, p, q)$$

gives this a precise formal referent.

It is the witness that the empirical record preserves a distinction that the algebraic regime has excluded.

The evidence knows the difference. The formal channel does not.

Verification is therefore not the reporting of pre-existing truth. It is the construction, under explicit structural conditions, of the proof obligations that make truth formally admissible in the first place.

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Data availability. This is a formal verification paper and contains no empirical datasets. The Rocq source files corresponding to the mechanised results reported in Sections 4–7, including the proofs of the Warrant Debt Theorem, the concrete rupture decomposition theorem for `SimpleSynergyCompose`, and the Non-Factorisation Theorem, are available in the canonical development file `bdgi_perturbation_proved.v` at `https://anonymous.4open.science/r/structural-admissibility-in-rocq-0C1C/`.

Code availability. The Rocq development is distributed under an open licence at the anonymous repository cited above and compiles against Rocq 8.18 or later.

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